2/ul/2026

function HILLCLIMBING(problem) returns a state which is a local maximum

current ← MAKE-NODE (problem. INITIAL-STATE)

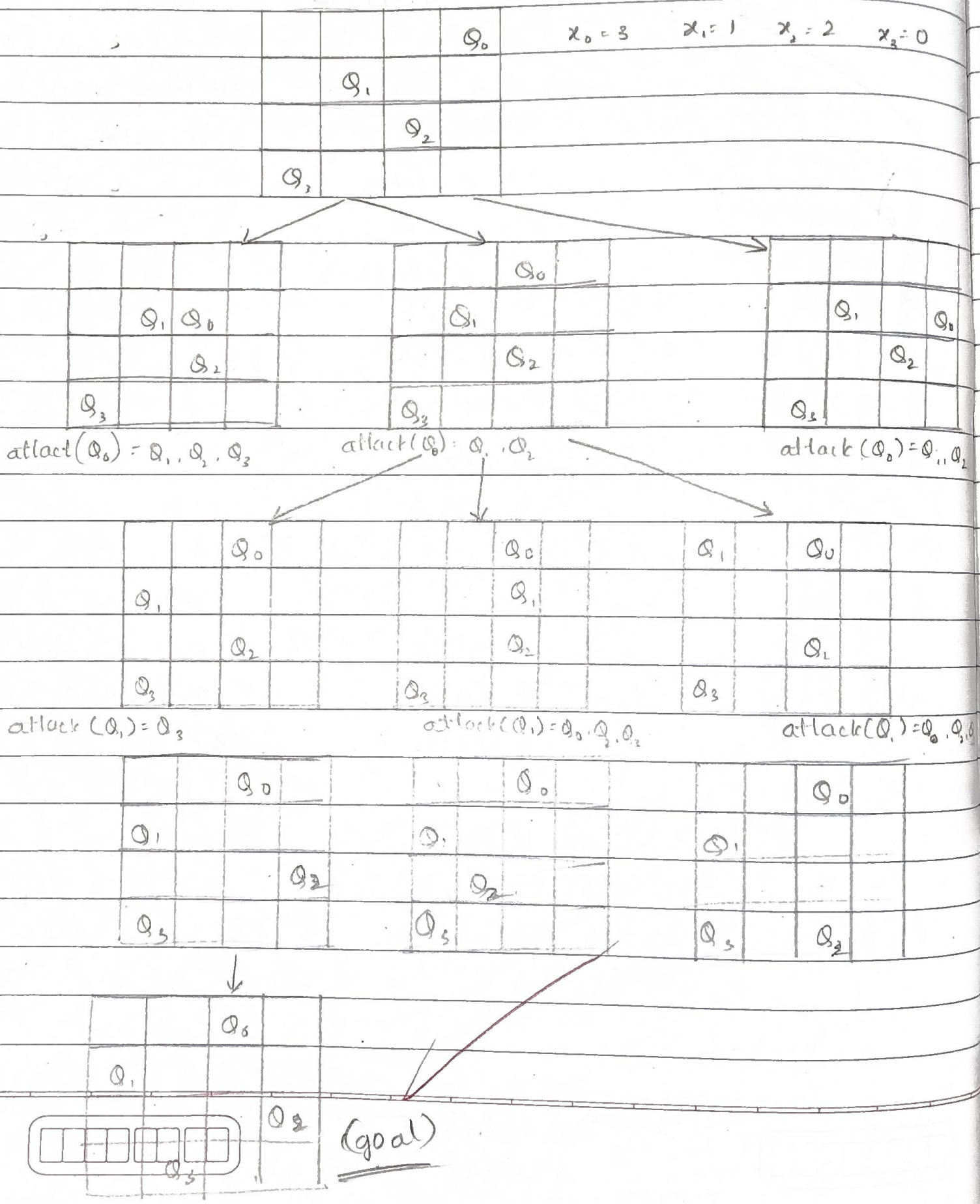
loop do

neighbor ← a highest valued successor of current

if neighbor.VALUE ≤ current.VALUE

then return current.STATE

current ← neighbor.



$(x_0, x_1, x_2, x_3) = (3, 1, 2, 0)$ cost = 2

① Swapped Queue | Neighbor State | cost

Swapped Queue	Neighbor State	cost	
x_0 x_1	(1, 3, 2, 0)	1	min=1
x_0 x_2	(2, 1, 3, 0)	1	
x_0 x_3	(0, 1, 2, 3)	6	
x_1 x_2	(3, 2, 1, 0)	6	
x_1 x_3	(3, 0, 2, 1)	1	
x_2 x_3	(3, 1, 0, 2)	1	

x_1 & x_2 can attack each other

② | | cost

x_0 x_1	(3, 1, 2, 0)	2
x_0 x_2	(2, 3, 1, 0)	2
x_0 x_3	(0, 3, 2, 1)	4
x_1 x_2	(1, 2, 3, 0)	4
x_1 x_3	(1, 0, 2, 3)	2
x_2 x_3	(1, 3, 0, 2)	0 (min)

